

The Growth of Executive Constraints

LLM Pipeline Codebook

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This codebook documents the dataset produced by the Gemini Batch API pipeline and open source models including Llama3.1-8B, Qwen2.5-32B, and Gemma3-27B. It covers every column in the canonical, full-dataset output files. The dataset can be found in the Google Drive folder.

Canonical run settings. The documented dataset is built WITHOUT self-consistency verification for Gemini pipeline and WITHOUT reasoning for all three open source models. Every indicator therefore contributes exactly two plain columns — `{ind}_prediction` and `{ind}_confidence` — except `constitution` (three extra document-level columns) and `elections` (a downstream pass-through rule).

1 Input Columns

The pipeline input is a leader-level table; the following columns from the original dataset are used.

Table 1: Input columns carried through the pipeline.

Column	Type	Meaning
<code>id</code>	string/int	Unique row identifier <i>[originated from <code>id</code> column]</i>
<code>slug_id</code>	string	Informative composite key, <code>polity_leader_years</code> ; not guaranteed unique in the raw source <i>[originated from <code>slug_id</code> column]</i>
<code>polity_name</code>	string	Name of the historical polity <i>[originated from <code>territorynamehistorical</code> column]</i>
<code>leader_name</code>	string	Name of the leader <i>[originated from <code>name_clean</code> column]</i>
<code>leader_first_year</code>	Int64 (nullable)	First year of tenure, negative = BCE <i>[originated from <code>entrydateyear</code> column]</i>
<code>leader_last_year</code>	Int64 (nullable)	Last year of tenure, negative = BCE <i>[originated from <code>exitdateyear</code> column]</i>

Missing `leader_first_year` / `leader_last_year` values are rendered as the literal string "unknown" inside prompts.

2 Indicator Definitions and Output Columns

Predictions are produced by three separate tasks that are later merged on `id`: `constitution` (its own dedicated prompt), `indicators` (all 17 non-constitution, non-elections indicators, combined into ONE prompt per leader-tenure), and `elections` (downstream, depends on `assembly_prediction`). Every column pair below is `{ind}_prediction` (the coded label) and `{ind}_confidence` (LLM self-reported confidence, integer 1–100).

2.1 Constitution (**constitution**)

Clarfication: Constitution is never part of the combined `indicators` prompt — it always runs as its own task/LLM call, using CoT and few-shot prompt.

Related columns:

Table 2: Output columns of the `constitution` task.

Column	Type	Meaning
<code>constitution_prediction</code>	int (0/1/2)	The highest type of written legal document that existed in the polity during this leader’s tenure. Codes: 0 = no written law 1 = code of law (governs subjects only) 2 = full constitution (written law + governance rules + limits on authority)
<code>constitution_confidence</code>	int (1–100)	LLM self-reported confidence score
<code>constitution_document_name</code>	string	Semicolon-separated name(s) of the referenced document(s), or "N/A"
<code>constitution_year</code>	string	Semicolon-separated adoption year(s), same order as <code>document_name</code>
<code>constitution_document_types</code>	string	Semicolon-separated type code(s) (1 or 2), same order as <code>document_name</code>

2.2 Indicators (combined **indicators** task)

Sovereign, federalism, the nine `checks_*` sub-indicators, collegiality, petition, assembly, entry, exit, and symbolism are NOT queried one prompt per indicator — they are combined into a single LLM call per leader-tenure.

The shared default system prompt opens with:

Sovereign (**sovereign**)

`sovereign_prediction` (int, 0–1):

- 0 = semi-sovereign
- 1 = sovereign domestic governance

`sovereign_confidence` (int, 1–100): LLM self-reported confidence score.

Federalism (**federalism**)

`federalism_prediction` (int, 0–1):

- 0 = unitary state
- 1 = federal — sovereignty divided between central and local units

`federalism_confidence` (int, 1–100): LLM self-reported confidence score.

Collegiality (collegiality)

`collegiality_prediction` (int, 0–1):

- 0 = single actor dominates decision-making
- 1 = decisions genuinely shared by a constituted body

`collegiality_confidence` (int, 1–100): LLM self-reported confidence score.

Petition (petition)

`petition_prediction` (int, 0–1):

- 0 = petitioning is rare, ineffective, or unrecorded
- 1 = petitions are a regular, institutionalized feature of political life

`petition_confidence` (int, 1–100): LLM self-reported confidence score.

Assembly (assembly)

`assembly_prediction` (int, 0–3):

- 0 = none
- 1 = council — small advisory council (institutionalized)
- 2 = legislature — policymaking or leadership-selection role
- 3 = popular assembly — most citizens, or a representative sample chosen by lot

`assembly_confidence` (int, 1–100): LLM self-reported confidence score.

Entry (entry)

`entry_prediction` (int, 0–10 or 99) — how the executive came to power:

- 0 = force (coup, rebellion, conquest)
- 1 = foreign power appointment
- 2 = ruling party appointment (one-party system)
- 3 = royal council appointment
- 4 = hereditary succession
- 5 = military appointment
- 6 = legislature appointment
- 7 = head of state appointment
- 8 = head of government appointment
- 9 = direct popular election
- 10 = other (e.g., clerical bodies such as the College of Cardinals)
- 99 = unknown

`entry_confidence` (int, 1–100): LLM self-reported confidence score.

Exit (exit)

`exit_prediction` (int, 0–14 or 99) — circumstances of the executive's departure:

- 0 = voluntary retirement or abdication (not due to ill health)
- 1 = other regular exit (term limits, electoral defeat)
- 2 = regular transition to another office
- 3 = died on campaign in civil war (disease or accident)
- 4 = died on campaign in foreign war (disease or accident)

- 5 = died of natural causes
- 6 = retired due to ill health
- 7 = suicide
- 8 = deposed by domestic actors
- 9 = assassinated or forced suicide
- 10 = killed in battle in civil war
- 11 = killed in battle in foreign war
- 12 = irregular transition to another office
- 13 = deposed by a foreign state
- 14 = still in office at end of observation period
- 99 = unknown

`exit_confidence` (int, 1–100): LLM self-reported confidence score.

Symbolism (`symbolism`)

`symbolism_prediction` (int, 0–3; non-monotonic):

- 0 = plain — little distinguishes the ruler from others in the realm
- 1 = decorated — impressive, but the officeholder is understood as mortal
- 2 = deified — the holder is regarded as divine or quasi-divine
- 3 = ceremonial — trappings so extensive they constrain the exercise of power

`symbolism_confidence` (int, 1–100): LLM self-reported confidence score.

2.3 Checks — Nine Independent Binary Sub-Indicators

The former 0–9 multi-select `checks` indicator was replaced with nine independent binary questions, one per actor, so that self-consistency agreement is well-defined per actor. “None” is not itself a question — it is the case where all nine columns equal 0. Each of the nine shares the same framing and coding scale (independence + de facto capacity to resist the executive; media, civil society, and ordinary citizens are NOT counted), reproduced below indicator-by-indicator like the other core indicators.

Checks: Local / Constituent Units (`checks_local`)

`checks_local_prediction` (int, 0–1):

- 0 = does not effectively check the executive
- 1 = effectively checks the executive

`checks_local_confidence` (int, 1–100): LLM self-reported confidence score.

Checks: Military (`checks_military`)

`checks_military_prediction` (int, 0–1):

- 0 = does not effectively check the executive
- 1 = effectively checks the executive

`checks_military_confidence` (int, 1–100): LLM self-reported confidence score.

Checks: Clergy (`checks_clergy`)

`checks_clergy_prediction` (int, 0–1):

- 0 = does not effectively check the executive

- 1 = effectively checks the executive

`checks_clergy_confidence` (int, 1–100): LLM self-reported confidence score.

Checks: Aristocracy (`checks_aristocracy`)

`checks_aristocracy_prediction` (int, 0–1):

- 0 = does not effectively check the executive
- 1 = effectively checks the executive

`checks_aristocracy_confidence` (int, 1–100): LLM self-reported confidence score.

Checks: Bourgeoisie (`checks_bourgeoisie`)

`checks_bourgeoisie_prediction` (int, 0–1):

- 0 = does not effectively check the executive
- 1 = effectively checks the executive

`checks_bourgeoisie_confidence` (int, 1–100): LLM self-reported confidence score.

Checks: Bureaucracy (`checks_bureaucracy`)

`checks_bureaucracy_prediction` (int, 0–1):

- 0 = does not effectively check the executive
- 1 = effectively checks the executive

`checks_bureaucracy_confidence` (int, 1–100): LLM self-reported confidence score.

Checks: Judiciary (`checks_judiciary`)

`checks_judiciary_prediction` (int, 0–1):

- 0 = does not effectively check the executive
- 1 = effectively checks the executive

`checks_judiciary_confidence` (int, 1–100): LLM self-reported confidence score.

Checks: Assembly (`checks_assembly`)

`checks_assembly_prediction` (int, 0–1):

- 0 = does not effectively check the executive
- 1 = effectively checks the executive

`checks_assembly_confidence` (int, 1–100): LLM self-reported confidence score.

Checks: Advisory Council (`checks_council`)

`checks_council_prediction` (int, 0–1):

- 0 = does not effectively check the executive
- 1 = effectively checks the executive

`checks_council_confidence` (int, 1–100): LLM self-reported confidence score.

2.4 Elections (**elections**) — Downstream Task

Run separately, after the main pipeline. Depends on `assembly_prediction`: rows where `assembly_prediction` $\neq 2$ receive a hard-coded pass-through label of "0" with NO API call; only rows with `assembly_prediction` = 2 get an LLM call.

Related columns:

Table 3: Output columns of the downstream elections task.

Column	Type	Meaning
<code>elections_prediction</code>	int (0/1/2)	0 = not elected, or pass-through when <code>assembly_prediction</code> $\neq 2$ 1 = elected, no organized factions 2 = competitive elections (organized factions or parties)
<code>elections_confidence</code>	int (1–100), null for pass-through rows	LLM self-reported confidence score

3 Dataset Statistics

This section reports label distributions and missingness for the current full-dataset snapshot of the Gemini dataset, `plt_constraints.csv` (135,672 rows, 51 columns).

3.1 Missing predictions in this dataset

The Gemini dataset is complete. All three tasks—`constitution`, `indicators`, and `elections`—now have *zero* missing predictions: every one of the 19 `{ind}_prediction` columns is non-null on all 135,672 rows. The earlier round of unparseable responses (14 rows on most indicators, 19 on `entry` and 18 on `exit`, caused by JSON truncation that nulled a whole call at once) was resubmitted through the retry-merge loop and has been filled in; the final row was repaired by hand from its raw response text.

The only column that is null by design is `elections_confidence`: it is empty on the 97,480 pass-through rows (`assembly_prediction` $\neq 2$) that never received an LLM call. Their `elections_prediction` is the hard-coded label 0.

Residual missingness now exists only in the open-weight runs documented in Section 5. The summary is below; Section 5.4 breaks it down by task and separates prediction-level from slot-level gaps.

Table 4: Missing predictions by pipeline, all 135,672 rows.

Pipeline	Missing rows	Detail
Gemini	0	Complete on all 19 indicators.
Llama-3.1-8B	0	Complete on all 19 indicators.

Table 4 (continued)

Pipeline	Missing rows	Detail
Qwen2.5-32B	31	The same 31 rows are null on all 17 indicators-task indicators (one unparseable combined call nulls the whole row); constitution and elections are complete.
Gemma-3-27B	0 (18 of 19)	Complete on the 17 indicators-task indicators and on elections, but carries <i>no</i> constitution column at all: that run did not complete (see Section 5.1 on the outlines backend).

3.2 Label distributions

Counts and percentages of total rows (135,672) for every category actually observed. There are no missing-prediction rows: every indicator sums to the full 135,672. Category meanings are defined in Section 2.

Table 5: Gemini label distributions, all 19 indicators ($N = 135,672$, zero missing).

Label	Count	% of rows
Constitution (constitution)		
0 — no written legal document	59,168	43.61%
1 — code of law (governs subjects only)	21,137	15.58%
2 — constitution (law + governance rules + limits)	55,367	40.81%
Sovereign (sovereign)		
0 — semi-sovereign (colony, protectorate, dependency)	58,025	42.77%
1 — sovereign over domestic affairs	77,647	57.23%
Federalism (federalism)		
0 — non-federal (unitary)	129,229	95.25%
1 — federal (incl. confederations, leagues)	6,443	4.75%
Collegiality (collegiality)		
0 — non-collegial (a single actor dominates)	117,224	86.40%
1 — collegial (power shared among co-equals)	18,448	13.60%
Petition (petition)		
0 — rare, ineffective, or unrecorded	81,180	59.84%
1 — a regular feature of political life	54,492	40.16%
Assembly (assembly)		
0 — none	55,060	40.58%
1 — council (small, advisory, institutionalized)	38,560	28.42%
2 — legislature (policymaking or selection role)	38,192	28.15%
3 — popular assembly (most citizens, or by lot)	3,860	2.85%
Entry (entry) — how the executive came to power		

Table 5 (continued)

Label	Count	% of rows
0 — force (coup, rebellion, conquest)	8,890	6.55%
1 — appointed by a foreign power	29,030	21.40%
2 — appointed by the ruling party	1,719	1.27%
3 — appointed by a royal council	6,125	4.51%
4 — hereditary succession	32,735	24.13%
5 — appointed by the military	528	0.39%
6 — appointed by the legislature	10,666	7.86%
7 — appointed by the head of state	19,125	14.10%
8 — appointed by the head of government	1,321	0.97%
9 — direct popular election	4,155	3.06%
10 — other (e.g. a clerical body)	7,632	5.63%
99 — unknown	13,746	10.13%
Exit (exit) — circumstances of departure		
0 — voluntary retirement or abdication	4,670	3.44%
1 — other regular exit (term limit, defeat)	43,197	31.84%
2 — regular transition to another office	6,308	4.65%
3 — died on campaign, civil war (disease/accident)	137	0.10%
4 — died on campaign, foreign war (disease/accident)	532	0.39%
5 — died of natural causes	34,735	25.60%
6 — retired due to ill health	888	0.65%
7 — suicide	224	0.17%
8 — deposed by domestic actors	8,558	6.31%
9 — assassinated or forced suicide	3,578	2.64%
10 — killed in battle, civil war	790	0.58%
11 — killed in battle, foreign war	2,079	1.53%
12 — irregular transition to another office	364	0.27%
13 — deposed by a foreign state	5,776	4.26%
14 — still in office	708	0.52%
99 — unknown	23,128	17.05%
Symbolism (symbolism) — non-monotonic; 3 is not “more” than 2		
0 — plain	35,917	26.47%
1 — decorated (officeholder understood as mortal)	74,889	55.20%
2 — deified (divine or quasi-divine status)	23,405	17.25%
3 — ceremonial (trappings constrain the ruler)	1,461	1.08%
Checks — nine binary sub-indicators. For each: 0 = absent, or present but unable to effectively resist the executive; 1 = independent and able to resist, at least on occasion.		
Checks — Local / Constituent Units (checks_local)		
0 — no effective check	109,327	80.58%
1 — effective check	26,345	19.42%
Checks — Military (checks_military)		

Table 5 (continued)

Label	Count	% of rows
0 — no effective check	124,984	92.12%
1 — effective check	10,688	7.88%
Checks — Clergy (<code>checks_clergy</code>)		
0 — no effective check	113,260	83.48%
1 — effective check	22,412	16.52%
Checks — Aristocracy (<code>checks_aristocracy</code>)		
0 — no effective check	97,697	72.01%
1 — effective check	37,975	27.99%
Checks — Bourgeoisie (<code>checks_bourgeoisie</code>)		
0 — no effective check	124,013	91.41%
1 — effective check	11,659	8.59%
Checks — Bureaucracy (<code>checks_bureaucracy</code>)		
0 — no effective check	129,945	95.78%
1 — effective check	5,727	4.22%
Checks — Judiciary (<code>checks_judiciary</code>)		
0 — no effective check	113,794	83.87%
1 — effective check	21,878	16.13%
Checks — Assembly (<code>checks_assembly</code>)		
0 — no effective check	99,567	73.39%
1 — effective check	36,105	26.61%
Checks — Advisory Council (<code>checks_council</code>)		
0 — no effective check	100,492	74.07%
1 — effective check	35,180	25.93%
Elections (<code>elections</code>)		
0 — pass-through (<code>assembly</code> $\neq$ 2, no LLM call)	97,480	71.85%
0 — LLM-coded: assembly exists but is not elected	12,273	9.05%
1 — elected, no organized factions	7,959	5.87%
2 — competitive (organized factions or parties)	17,960	13.24%

The `elections` block separates the two kinds of zero: 97,480 rows are labelled 0 by the pass-through rule without any model call, and only the remaining 38,192 rows (`assembly_prediction` = 2) were actually judged by the model, of which 12,273 were coded 0. Any analysis of `elections` should keep these two groups apart.

3.3 Constitution document-metadata missingness

Within the `constitution` task, the three document-level columns — `constitution_document_name`, `constitution_document_types`, and `constitution_year` — are missing at very different rates once conditioned on `constitution_prediction`:

Table 6: Constitution document-metadata missingness

Prediction	Rows	document_name missing	document_types missing	constitution_year missing
0 (no written law)	59,168	59,166 (100.00%)	59,166 (100.00%)	59,166 (100.00%)
1 (code of law)	21,137	5 (0.02%)	6 (0.03%)	1,519 (7.19%)
2 (full constitution)	55,367	0 (0.00%)	0 (0.00%)	777 (1.40%)

At prediction = 0 all three columns are missing together, as expected (no document to name, type, or date). At prediction = 1 or 2, however, `constitution_year` is missing far more often than `document_name`/`document_types` — roughly 300–600× more rows. This is a deliberate consequence of the prompt wording (Section 2.1): the model is instructed to give the adoption year *as an exact integer only, or "N/A" if uncertain — no approximations* (“c.”, “circa”, “~”), with no equivalent exactness requirement on the document’s name or type. In practice the model is much more often able to identify and classify a legal document than to pin down its precise adoption year, especially for premodern or poorly-documented polities; the asymmetry reflects the model honestly reporting that uncertainty rather than a parsing defect. (A separate check found 2 rows with prediction = 0 yet non-null document metadata — an internally inconsistent LLM output, negligible at 0.001% of rows.)

4 Related Files

4.1 Output files

Table 7: Output files written by the Gemini batch pipeline.

File	Description
<code>plt_constraints.csv</code>	Tabular prediction output. All original input columns are preserved, with the new <code>{ind}_prediction/</code> <code>{ind}_confidence</code> (and constitution’s three extra) columns appended.
<code>{task}_provenance.json</code>	Per-run provenance record written by the batch runner: <code>model</code> , <code>run_timestamp</code> , <code>n_samples</code> , <code>input_jsonl</code> (paths of the request file(s) submitted), <code>total_requests</code> , <code>success_count</code> , <code>response_failed_cids</code>
<code>{task}_failed_requests.jsonl</code>	Only written when at least one row failed. Contains exactly the failed rows’ requests, ready to resubmit as-is to <code>jsonl_batch_runner.py</code> .

4.2 Chaining the three tasks

1. `constitution build + run` from the raw input → `plt_constitution.csv`
2. `indicators build + run` from the raw input → `plt_indicators.csv`
3. `elections build FROM plt_indicators.csv` (requires its `assembly_prediction` column), run with `--dataset plt_indicators.csv` → final merged dataset

5 Open-Weight Model Runs

Everything above documents the Gemini Batch API dataset. The same three tasks were additionally run on three open-weight models served locally with vLLM on TACC Lonestar6. These runs share the pipeline’s prompts, label sets, coding rules and column-naming conventions; what differs is the sampling regime (self-consistency is ON) and the availability of token-level probabilities, which a hosted API does not expose. This section documents only those differences.

5.1 Models and run configuration

Table 8: Open-weight models and run configuration (vLLM on TACC Lonestar6).

	Llama	Gemma	Qwen
Checkpoint	Llama-3.1-8B-Instruct	Gemma-3-27b-it	Qwen2.5-32B-Instruct
Parameters	8B	27B	32B
Tensor parallel	1	2	2
Throughput	0.29 s/row	≈1.6 s/row	1.59 s/row
Full-dataset time	≈11 h, one job	≈60 h, 4 shards	≈60 h

Throughput is per dataset row and already includes all three self-consistency samples. The two larger models require tensor parallelism because neither fits in a single 40 GB accelerator at `bfloat16`; the 8B model does, which is why it completes within one scheduling window while the others must be sharded or chained.

Settings held constant across all three models. Identical request files (the serialised prompts are model-independent; each model’s chat template is applied at run time from its own checkpoint), `max_model_len` 8192, `max_tokens` 2048, `dtype bfloat16`, `seed` 42, `top-20` token logprobs retained, and two additional self-consistency samples, giving **three votes per row**. Sampling temperature and nucleus parameter are carried in the request file and are therefore identical across models. Reasoning text is disabled, so no `{ind}_reasoning` columns are emitted.

No quantisation. All three run at `bfloat16`. Quantisation would perturb the logits from which the probability columns in Section 5.2 are derived, so it is never applied.

5.2 Columns not present in the Gemini dataset

Two families of columns appear only in the open-weight outputs.

(a) Self-consistency slots. The Gemini dataset is a single pass, so each indicator contributes `{ind}_prediction` and `{ind}_confidence`. The open-weight runs draw three independent samples per row, numbered `SC1–SC3`, and reduce them by majority vote.

Table 9: Self-consistency columns present only in the open-weight outputs.

Column	Meaning
<code>{ind}_SC1..SC3</code>	The label produced by each individual sample. SC1 is the initial call, SC2–SC3 the two additional samples. Null when that sample’s output could not be parsed.
<code>{ind}_prediction</code>	Majority vote across the three slots. Present even when a slot is null, in which case the vote is taken over the surviving slots — see the caution below.
<code>{ind}_agreement</code>	The winning label’s vote count divided by the number of <i>non-null</i> slots, in $[0, 1]$.
<code>{ind}_uncertainty</code>	none if unanimous, low if the winner holds a majority, high if all slots differ (in which case the prediction falls back to SC1).
<code>{ind}_confidence_ SC1..SC3</code>	Per-sample self-reported confidence, integer 1–100. Replaces the single <code>{ind}_confidence</code> column of the Gemini dataset.

Caution on the agreement denominator: `{ind}_agreement` is computed over non-null slots only. A row with one unparseable slot therefore reports a ratio out of two rather than three, and nothing in the column itself signals this. Any analysis treating agreement as comparable across rows should first check the `{ind}_SCn` columns for nulls and exclude or separately flag those rows.

(b) Token-level probability columns. For every indicator and every self-consistency slot the runner records the model’s probability distribution over the *legal labels for that indicator* at the position where the label token is emitted. Four columns per indicator per slot:

Table 10: Token-level probability columns present only in the open-weight outputs.

Column	Meaning
<code>{ind}_label_dist_SCn</code>	JSON object mapping each legal label to its probability. Keys are the label values as they appear in <code>{ind}_prediction</code> ; values sum to 1.
<code>{ind}_prob_SCn</code>	Probability mass assigned to the label the model actually emitted — the entry of <code>label_dist</code> selected in that slot.
<code>{ind}_logprob_SCn</code>	Natural logarithm of the preceding column. Redundant with it by construction; provided because the log scale is the convenient one for likelihood-style aggregation.
<code>{ind}_entropy_SCn</code>	Shannon entropy of the full <code>label_dist</code> , in nats. Zero means all mass on one label; the maximum is $\ln K$ for an indicator with K legal labels.

How these are computed. The generator is asked for the top 20 token logprobs at each position. At the label position those logprobs are restricted to the tokens corresponding to legal labels for that indicator, exponentiated, and renormalised to sum to 1; the result is `label_dist`. Mass on tokens outside the

legal set is discarded rather than redistributed, so these are *conditional* probabilities given that a legal label was produced. Because constrained decoding already forbids illegal tokens, the discarded mass is normally negligible.

Three columns, one quantity. `prob`, `logprob` and `entropy` are not three independent measurements. The first two are the same number on different scales; the third summarises the whole distribution rather than the selected label. A row can show high `prob` together with high `entropy` only when the remaining mass is spread thinly over many labels.

Single-token label requirement. Per-label probability and entropy are well defined only if each legal label occupies exactly one token. The labels for `entry` (0–10, 99) and `exit` (0–14, 99) do not satisfy this, because multi-digit values tokenise into more than one piece. These two indicators are therefore *aliased* at prompt-construction time: `entry` uses A–K for 0–10 and Z for 99; `exit` uses A–O for 0–14 and Z for 99. The model emits letters, and the runner maps them back to the integer codes before anything is written out. **All published columns — including the keys inside `label_dist` — use the integer codes documented earlier in this codebook.** The letters never appear in the output.

5.3 Output file inventory

Each model produces one file per task, all 135,672 rows, aligned row-for-row with the input on `id`.

Table 11: Open-weight output file inventory (per model).

File	Columns	Contents
<code>exp_{model}_indicators.csv</code>	363	6 input columns + 17 indicators × 21 columns each: 3 SC slots × {label, confidence, label_dist, prob, logprob, entropy}, plus prediction, agreement, uncertainty.
<code>exp_{model}_constitution.csv</code>	33	6 input columns + 1 indicator × 27 columns: 3 SC slots × {label, document_name, year, document_types, label_dist, prob, logprob, entropy}, plus prediction, agreement, uncertainty. Constitution carries document metadata per slot <i>instead of</i> a confidence slot.
<code>exp_{model}_elections.csv</code>	384	The indicators frame plus the elections columns. Rows whose assembly prediction is not 2 pass through with <code>elections = 0</code> and no model call, so their SC and probability columns are empty <i>by design</i> , not by failure.
<code>plt_constraints_{model}.csv</code>	411	The final per-model file. The elections frame (384) merged with the constitution columns (27). This is the one to analyse; the three files above are intermediates.

Each file supersedes the one above it: a completed `elections.csv` contains every indicators column as well, and `plt_constraints_{model}.csv` in turn contains everything in `elections.csv` plus the constitution columns.

At the time of writing, `plt_constraints_llama8b.csv` and `plt_constraints_qwen32b.csv` exist at the full 411 columns. Gemma has only `gemma27b_elections.csv` (384 columns, 18 pre-

dictions): its constitution run is outstanding, so Gemma carries no `constitution` column anywhere and is absent from the constitution row of the comparison table below.

A sidecar `{stem}_raw.jsonl` accompanies each run, holding one line per (row, sample) with the raw generated text, the finish reason and the token logprobs. It is the source of truth from which the CSV is rebuilt. It is appended to rather than overwritten, so re-running selected rows leaves the superseded lines in place; aggregation keeps the last write for each (row, sample) key.

5.4 Missingness in the open-weight runs

Two different things can be missing here, and only the first is visible in the prediction columns. **Prediction-level** missingness means the merged `{ind}_prediction` is null: no slot parsed, so no vote could be taken. **Slot-level** missingness means one of `SC1–SC3` failed to parse while the remaining slots still produced a majority — the prediction is non-null and looks perfectly healthy, but it rests on two votes rather than three and its `{ind}_agreement` is a ratio out of two. This is exactly the denominator problem flagged in Section 5.2; the counts below are how large it actually is.

Table 12: Missingness in the open-weight runs, by model and task.

Model	Task	Rows scored	Missing predictions	Rows with ≥ 1 unparsed slot
Llama	constitution	135,672	0	68
	indicators	135,672	0	0
	elections	31,579	0	0
Qwen	constitution	135,672	0	32
	indicators	135,672	31	85
	elections	11,251	0	0
Gemma	constitution	<i>not run — no column exists</i>		
	indicators	135,672	0	0
	elections	26,152	0	0

For `constitution` and `indicators` every model was asked all 135,672 rows. For `elections` the gate admits only rows the *same model* coded `assembly = 2`, so “rows scored” differs by model by construction and is not a sign of failure; the remaining rows carry the pass-through label 0 with empty `SC` and probability columns, again by design.

Read together with the Gemini figures in Section 3.2: at the prediction level only Qwen’s `indicators` task is incomplete, and there the same 31 rows are null across all 17 indicators at once — the signature of one combined call whose JSON never parsed, not of 17 independent failures. Slot-level gaps are concentrated in `constitution` (68 rows for Llama, 32 for Qwen), which is the only task with free-string fields and therefore the only one whose output can be truncated mid-string; Qwen’s 85 slot-level rows in `indicators` include its 31 prediction-level rows.

6 Cross-Model Comparison

The table below compares each open-weight model against the Gemini dataset documented in the preceding sections. Agreement is the proportion of rows on which the two pipelines emit the same la-

bel, computed only where both produced a non-null prediction. For the open-weight models, `{ind}_prediction` is the **majority vote over the three self-consistency samples**, not any single sample.

The Qwen elections row of both tables below is under correction — see the boxed note in Section 5.4; that column is very probably Gemma output and will be regenerated.

$N = 135,672$ throughout, except the Qwen indicator rows where $N = 135,641$ (31 rows carry no Qwen prediction). The Gemini side is now non-null everywhere, so N is bounded only by the open-weight column. Gemma has no `constitution` column: that run did not complete, for the reason given in the previous section.

Table 13: Open-weight agreement with Gemini, by indicator.

Indicator	Llama	Qwen	Gemma
sovereign	0.573	0.798	0.639
federalism	0.915	0.949	0.906
checks_local	0.789	0.809	0.696
checks_military	0.918	0.911	0.292
checks_clergy	0.840	0.844	0.710
checks_aristocracy	0.682	0.734	0.526
checks_bourgeoisie	0.873	0.911	0.886
checks_bureaucracy	0.928	0.950	0.690
checks_judiciary	0.857	0.846	0.860
checks_assembly	0.790	0.769	0.825
checks_council	0.741	0.726	0.435
collegiality	0.872	0.875	0.623
petition	0.627	0.598	0.696
assembly	0.433	0.463	0.484
entry	0.290	0.291	0.370
exit	0.318	0.423	0.462
symbolism	0.376	0.553	0.494
constitution	0.334	0.508	—
elections	0.847	0.840	0.814

Reading these numbers. Raw agreement is not comparable across indicators, for two reasons that pull in opposite directions. First, it rises mechanically with how skewed an indicator’s label distribution is: `federalism` exceeds 0.9 for all three models largely because everyone answers 0 on the overwhelming majority of rows, so most of that figure is concordance any two coders with the same marginal would reach without agreeing about anything in particular. Second, it falls mechanically with the number of categories, which is why `entry` (12 labels) and `exit` (16) sit lowest. Read each figure against that indicator’s label distribution in Section 3.2 rather than against the other rows of this table.

Comparison *across* the three columns is the more useful direction, because category count and marginal skew are held fixed within a row. Qwen is closest to Gemini on most indicators, and markedly so on `sovereign` (0.798 against Llama’s 0.573) and `symbolism` (0.553 against 0.376). Gemma is the outlier in both directions: it alone falls below 0.5 on `checks_military` (0.292, against roughly 0.91 for the other two) and on `checks_council` (0.435), yet it leads on `entry`, `exit` and `petition`. A single indicator collapsing to 0.292 while its neighbours hold above 0.9 is more consistent with a

systematic labelling difference — one model reading that indicator’s definition differently — than with diffuse noise. `checks_military` is worth inspecting directly before the Gemma column is used.

The elections row is not what it appears. Its high values are largely definitional rather than substantive. Rows whose `assembly` prediction is not 2 receive `elections` = 0 by rule with no model call, and that covers 97,480 of the Gemini rows (a further 12,273 model-coded rows are also labelled 0, giving 109,753 zeros in total). Both pipelines therefore assign 0 to most rows without either having judged anything. Restricting to rows where *both* pipelines actually invoked the model gives a different picture:

Table 14: `elections` agreement, restricted to rows both pipelines actually scored.

Model	Rows both scored	Agreement (scored)	Agreement (all rows)
Llama	3,298	0.857	0.847
Qwen	6,763	0.761	0.840
Gemma	17,424	0.531	0.814

Note the row counts. Gemini assigns `assembly` = 2 to 38,192 rows, but only 3,298 of those are also gated in by Llama — an overlap of 8.6%. Because `assembly` itself agrees at only 0.43–0.48, each pipeline’s elections gate selects a substantially different set of leader-tenures. **The elections columns of two pipelines are therefore not directly comparable:** they answer the same question about different populations. Any analysis using `elections` should either restrict to rows gated in by both pipelines, or state which pipeline’s `assembly` defined the sample.

What this does and does not establish. These figures measure *inter-model* agreement, not accuracy. Three indicators (`sovereign`, `collegiality`, `assembly`) have human-coded ground truth against which any of these models can be scored directly, and that comparison is the one that speaks to validity. Low agreement between two models is consistent with one being accurate and the other not, as well as with both being unreliable; it does not by itself adjudicate between those cases. It does mean the datasets are not interchangeable, and that any downstream result should state which model produced the labels it rests on.

Comparability caveats. The pipelines differ in more than the model. The Gemini dataset is a single pass; every open-weight column is a majority vote over three sampled generations. Sampling at nonzero temperature and then voting is not the same estimator as one decode, and part of the disagreement above is attributable to that difference rather than to the models themselves. A like-for-like comparison would either take `{ind}_SC1` alone or re-run Gemini under matched sampling.